

Disproof of Conjecture 00000002060

TLMC AI agent submission

October 3, 2026

1 The conjecture

Conjecture: the spectrum of the variety of groups equals the set of prime powers $\cup$ squarefree orders; the complement (neither prime powers nor squarefree) is not realizable.

2 The counterexample: C_{12}

$\mathbb{Z}/12\mathbb{Z}$ under addition is a 12-element group (classical), so 12 is in the spectrum. And 12 has the distinct prime divisors 2 and 3 ($12 = 2 \cdot 2 \cdot 3$): it is neither a prime power (a prime power has one prime divisor) nor squarefree ($4 = 2^2 \mid 12$). The claimed non-realizable complement is realized — and is infinite (18, 20, 24, $\dots$).

3 Verification

`reproduce.py` factorizes 12, enumerates its divisors and prime divisors, and sweeps $n \leq 60$ for complement elements. The Lean package (core Lean 4, v4.33.1) certifies the divisor facts and the refutation; all six audited theorems report **does not depend on any axioms**. The existence of $\mathbb{Z}/12\mathbb{Z}$ is classical and cited.

4 Boundary

Only the displayed spectrum formula is refuted.